

DSA02-CO2-AT3

1. An image appears too bright

Answer

Concept

A bright image has very high pixel intensity values, making details difficult to see because many pixels are close to white.

Suitable Grey Level Transformation

Negative (Brightness Reduction) / Intensity Scaling (Linear Transformation)

A **linear grey level transformation** can reduce the brightness by multiplying each pixel by a constant less than 1.

Formula:

$$s = c \times r, (0 < c < 1)$$

where:

- r = Original pixel value
- s = New pixel value
- c = Brightness reduction factor

Application Process

1. Read the image.
2. Multiply every pixel intensity by a value less than 1.
3. Generate the enhanced image.

Expected Outcome

- Brightness decreases.
- Hidden details become visible.
- Image appears more natural.

Justification

This transformation is suitable because it reduces excessive brightness while preserving the overall image information.

2. A dark image lacks detail

Answer

Concept

A dark image has very low pixel intensity values, making objects difficult to identify.

Suitable Grey Level Transformation

Log Transformation

Formula

$$s = c \log(1+r) \quad s = c \log(1+r)$$

Working

- Expands low-intensity values.
- Compresses high-intensity values.
- Brightens dark regions.

Application Process

1. Apply the log transformation to each pixel.
2. Increase the visibility of dark pixels.
3. Produce a brighter image.

Expected Outcome

- Dark regions become brighter.
- Hidden details become visible.
- Overall image quality improves.

Justification

Log transformation is ideal because it enhances low-intensity pixels much more than high-intensity pixels.

3. An image has poor contrast

Answer

Concept

Poor contrast means the intensity values are concentrated within a small range.

Suitable Technique

Histogram Equalization

Working

- Calculates the image histogram.
- Computes the cumulative distribution function (CDF).
- Redistributes intensity values over the full intensity range.

Application Process

1. Find the histogram.
2. Calculate cumulative frequencies.
3. Map old intensities to new values.

Expected Outcome

- Better contrast.
- Clearer object boundaries.
- Improved image visibility.

Justification

Histogram Equalization spreads pixel values evenly, making details easier to observe.

4. A low-contrast image requires uniform intensity distribution

Answer

Concept

The image has uneven intensity distribution, causing poor visibility.

Suitable Method

Histogram Equalization

Working Process

- Computes the histogram.
- Uses cumulative probability.
- Redistributes pixel intensities uniformly.

Effect on Intensity Levels

- Dark pixels become darker where needed.
- Bright pixels become brighter where needed.
- Overall intensity becomes more evenly distributed.

Expected Result

- Improved contrast.
- Better visibility.
- Easier image interpretation.

Justification

Histogram Equalization is specifically designed to produce a more uniform intensity distribution.

5. An image shows uneven brightness across regions

Answer

Concept

Different parts of the image have different brightness because of non-uniform lighting.

Suitable Enhancement Technique

Homomorphic Filtering

Working

- Separates illumination and reflectance.
- Reduces illumination variations.
- Enhances reflectance (actual object information).

Application Process

1. Convert image to logarithmic form.

2. Apply Fourier Transform.
3. Apply Homomorphic Filter.
4. Perform Inverse Fourier Transform.

Expected Outcome

- Uniform brightness.
- Better contrast.
- Improved image appearance.

Justification

Homomorphic Filtering corrects uneven illumination while preserving important image details.

6. A noisy image contains random intensity variations

Answer

Concept

Noise introduces unwanted random pixel values into the image.

Suitable Smoothing Filter

Median Filter (*for impulse/salt-and-pepper noise*)

or

Mean Filter (*for general random noise*)

Working

The Median Filter replaces each pixel with the median value of its neighboring pixels.

Application Process

1. Select a 3×3 window.
2. Arrange neighboring values.
3. Replace the center pixel with the median value.

Expected Outcome

- Noise is removed.
- Edges remain sharp.

- Image quality improves.

Justification

Median filtering removes noise without significantly blurring important image details.

7. An image requires noise reduction without significant loss of details

Answer

Concept

Noise should be removed while preserving edges and fine details.

Suitable Filtering Approach

Median Filter

Working

- Removes isolated noisy pixels.
- Preserves edge information.
- Does not average neighboring pixels.

Application Process

1. Choose neighborhood.
2. Compute median.
3. Replace noisy pixel.

Expected Result

- Noise reduction.
- Sharp edges.
- Better image quality.

Justification

Median filtering provides an excellent balance between noise removal and detail preservation.

8. An image appears blurred after smoothing

Answer

Concept

Excessive smoothing removes high-frequency information, causing blurred edges.

Suitable Sharpening Method

Laplacian Sharpening

Working Principle

The Laplacian operator detects rapid intensity changes and enhances image edges.

Common Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Application Process

1. Apply the Laplacian mask.
2. Detect edges.
3. Add the result to the original image.

Expected Result

- Sharper edges.
- Improved object boundaries.
- Better image clarity.

Justification

Laplacian sharpening restores edge information lost during smoothing.

9. Fine details in an image are not clearly visible

Answer

Concept

Fine details correspond to high-frequency components of an image.

Suitable Filtering Technique

High-Pass Filter

Working

- Removes low-frequency information.

- Preserves high-frequency components.
- Enhances edges and textures.

Effect on High-Frequency Components

- Increases edge sharpness.
- Highlights fine structures.
- Improves detail visibility.

Expected Result

- Clearer fine details.
- Better texture representation.
- Enhanced image quality.

Justification

High-pass filtering is specifically designed to enhance high-frequency image information.

10. A noisy image must be preprocessed before further analysis

Answer

Concept

Noise can reduce the accuracy of image segmentation, feature extraction, and object detection.

Suitable Spatial Filtering Method

Median Filter

(Mean Filter may be used for Gaussian noise.)

Working

The filter processes each pixel using information from neighboring pixels and replaces noisy values with more representative values.

Application Process

1. Select a neighborhood (3×3).
2. Compute the median.
3. Replace the center pixel.

4. Repeat for the entire image.

Expected Outcome

- Reduced noise.
- Better image quality.
- Improved segmentation and feature extraction.

Justification

Median filtering effectively removes noise while preserving edges, making it an excellent preprocessing technique before further image analysis.